University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2021, Part 1

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let G be a cyclic group. Prove that every subgroup of G is cyclic. (Be sure to consider infinite cyclic groups as well as finite ones.)

2. Let G, H be finite groups with orders m, n.

(a) Prove that if m and n are coprime then

$$\mathrm{Aut}(G \times H) \cong \mathrm{Aut}(G) \times \mathrm{Aut}(H).$$

(b) Give an example where m and n are not relatively prime and

$$\mathrm{Aut}(G \times H) \ncong \mathrm{Aut}(G) \times \mathrm{Aut}(H).$$

Justify your claims.

3. This problem has two parts.

(a) Let G be a finite group of order n and let p be the smallest prime which divides n. Prove that if H is a subgroup of G of index p then H is normal in G.

(b) Give an example of a finite group G and a subgroup $H \leq G$ such that H is not normal in G and the index $|G : H|$ is prime.

4. Let p, q, r be primes such that $p < q < r$ and let G be a group of order pqr. Prove that G has a normal Sylow subgroup.

5. Prove that the alternating group A_5 is simple. Prove that A_4 is solvable.

6. Show that there are four different homomorphisms $\phi : \mathbb{Z}_2 \to \mathrm{Aut}(\mathbb{Z}_8)$, and prove that the corresponding semidirect products $\mathbb{Z}_8 \rtimes_\phi \mathbb{Z}_2$ are pairwise nonisomorphic. (Hint: Consider the number of elements of order 2.)